Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI for Network Security

TGS Ref No: TGS-2024051414

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.1

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	29 August 2026	Initial release of the AI for Network Security lesson plan — 2 days, 8 instructional hours per day, 4 topics aligned to LO1-LO4 and TSC ICT-DIT-4024-1.1, 12 hands-on AI-assisted labs, and the WA-SAQ + Case Study assessment on Day 2.	Dr. Alfred Ang
1.1	29 August 2026	Quality-audit revision. Process-map stage labels are now authored short names in the single source (never derived by truncation), with a build-time guard; procedure-slide headings use one uniform size per slide; the Support slide became a tile grid; the day schedule moved to a 9:30 am start with exactly 8 instructional hours per day (lunch and tea breaks excluded, enforced by assertion); the Case Study is 1 hour, matching the paper and the deck.	Dr. Alfred Ang

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Course Information

Course Title	AI for Network Security
WSQ Course Reference	TGS-2024051414
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 instructional hours per day (16 hours)
Daily Timing	Day 1: 9:30 am – 7:00 pm · Day 2: 9:30 am – 7:00 pm (1-hour lunch and two 15-minute tea breaks, excluded from the 8 instructional hours)
Mode	Instructor-led, hands-on AI-assisted network security labs against authorised lab targets
Skills Framework	TSC: Network Security (ICT-DIT-4024-1.1)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Identify and assess network threats and vulnerabilities by analysing ICT configurations.
- LO2: Evaluate organisational assets and develop risk management plans to address network infrastructure weaknesses effectively.
- LO3: Design and implement security measures for network components by managing security phases.
- LO4: Document security incidents and apply incident response configurations to manage risks based on established procedures.

Assessment

- Written Assessment — Short-Answer Questions (WA-SAQ) · 1 hour · open book. Assesses the underpinning knowledge K1–K4.
- Case Study (CS) · 1 hour · open book. Assesses the abilities A1–A9 against a realistic organisational scenario.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 5:00 pm.
- Knowledge (K1-K4) is assessed by the WA-SAQ; abilities (A1-A9) are assessed by the Case Study.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

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Course Schedule

Day 1 — AI-Assisted Threat & Vulnerability Analysis · AI-Powered Risk Assessment and Management Planning

Time	Duration	Topic / Activity	Slides
9:30–10:00	30 min	Welcome, course introduction, ground rules, learning outcomes and mandatory digital attendance (AM)	1-19
10:00–11:15	75 min	**Core concepts — network security and the CIA triad; why signature-based detection fails against zero-day, polymorphic and Living-off-the-Land attacks; where AI is applied in network security (concepts + demo)**	20-31
11:15–11:30	15 min	Tea break	
11:30–13:30	120 min	Topic 1 — AI-Assisted Network Threat and Vulnerability Analysis: ICT networks and configuration (K1), the threat landscape (K2), pattern-of-life baselines, NDR and AI vulnerability prioritisation (concepts). Hands-on: Lab 1: Map the ICT Network and Analyse Its Configuration; Lab 2: Baseline Normal Traffic and Detect an Anomaly with AI	32-50
13:30–14:30	60 min	Lunch break	
14:30–16:15	105 min	Hands-on: Lab 3: AI-Assisted Vulnerability Analysis and Prioritisation. Topic 2 — AI-Powered Network Risk Assessment and Management Planning: asset discovery and classification, CIA business impact (concepts)	51-65
16:15–16:30	15 min	Tea break	

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16:30–19:00	150 min	Topic 2 continued — risk scoring on the 5x5 matrix, the four treatments, NIST CSF 2.0 and the NIST AI RMF (concepts). Hands-on: Lab 4: Build an Asset Register and Classify What Needs Protecting; Lab 5: AI-Assisted Risk Scoring and the Risk Register; Lab 6: Develop the Risk Management Plan. Day 1 recap, Q&A and PM digital attendance	66-81

Total instructional time: 480 minutes (8 hours), excluding the 1-hour lunch and the two 15-minute tea breaks.

Day 2 — Designing & Implementing AI-Enhanced Security Measures · AI-Assisted Incident Documentation and Response

Time	Duration	Topic / Activity	Slides
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)	
9:45–11:15	90 min	**Topic 3 — Designing and Implementing AI-Enhanced Network Security Measures: defence in depth and the three control categories; the planning, building and management phases; segmentation, zero trust and access control (K3, concepts + demo)**	82-94
11:15–11:30	15 min	Tea break	
11:30–13:30	120 min	Topic 3 continued — AI-enhanced firewalls, IDS/IPS, UEBA and adaptive access control; securing the AI itself (NIST AI 100-2 attack taxonomy, NSA/CISA deployment guidance, MITRE ATLAS). Hands-on: Lab 7: Design the Layered Architecture Across the Three Security Phases; Lab 8: Implement AI-Enhanced Controls —	95-111

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		Firewall Policy, IDS and Adaptive Access; Lab 9: Secure the AI Itself — Adversarial ML and the NSA/CISA Deployment Controls	
13:30–14:30	60 min	Lunch break	
14:30–16:00	90 min	Topic 4 — AI-Assisted Security Incident Documentation and Response: logging and audit design, SIEM/SOAR, AI alert triage, the NIST SP 800-61 lifecycle and incident documentation standards (K4, concepts). Hands-on: Lab 10: Centralise Logs and Triage Alerts with AI; Lab 11: Document a Security Incident to the Standard	112-134
16:00–16:15	15 min	Tea break	
16:15–16:45	30 min	Hands-on: Lab 12: Implement the Incident Response Configuration and Playbook. Course recap, TRAQOM survey and assessment digital attendance	135, 142-147
16:45–17:00	15 min	**Briefing for Assessment**	148
17:00–18:00	60 min	**Written Assessment — Short-Answer Questions (WA-SAQ), 1 hour, open book. Assesses the underpinning knowledge K1-K4**	148-149
18:00–19:00	60 min	**Case Study (CS), 1 hour, open book. Assesses the abilities A1-A9 against a realistic organisational scenario**	148-151

Total instructional time: 480 minutes (8 hours), excluding the 1-hour lunch and the two 15-minute tea breaks.

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Lab Reference (aligned to learning outcomes and TSC statements)

Topic / Learning unit	Weighting	K / A assessed	Labs	Slides
Topic 01: AI-Assisted Network Threat and Vulnerability Analysis	25%	K1 | A1, A2	Lab 1, Lab 2, Lab 3	Slides 41-51
Topic 02: AI-Powered Network Risk Assessment and Management Planning	25%	K2 | A3, A4	Lab 4, Lab 5, Lab 6	Slides 66-76
Topic 03: Designing and Implementing AI-Enhanced Network Security Measures	30%	K3 | A5, A6, A7	Lab 7, Lab 8, Lab 9	Slides 95-105
Topic 04: AI-Assisted Security Incident Documentation and Response	20%	K4 | A8, A9	Lab 10, Lab 11, Lab 12	Slides 123-135

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